SUPPLEMENTARY MATERIALS: A CONTINUOUS-TIME ACTOR-CRITIC FRAMEWORK FOR JUMP-DIFFUSION CONTROLS WITH NORMALIZING FLOWS*

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This document supplements the main manuscript, " A Continuous-Time Actor-Critic Framework for Jump-Diffusion Controls with Normalizing Flows" by collecting theoretical statements, technical derivations and additional details that support the analysis and numerical results in the paper.

S.1. Proof of Theoretical Results.

S.1.1. Derivation of the q-Function (3.8). We derive (3.7), which leads to the definition of the little q-function (3.8), via a short-time expansion of Q_{δ_t} defined in (3.6). Fix $\delta_t > 0, t \geq 0$, $\boldsymbol{x} \in \mathbb{R}^d$, and $\boldsymbol{u} \in \mathcal{A}$, and recall $(\boldsymbol{X}_s^{\boldsymbol{u}})_{s\geq t}$ from Section 3.2.1. By the tower property and $e^{-\beta(s-t)} = e^{-\beta\delta_t}\, e^{-\beta(s-(t+\delta_t))}$, Q_{δ_t} can be rewritten as

$$
\begin{aligned}
Q_{\delta_t}(t,\boldsymbol{x},\boldsymbol{u};\pi) = \mathbb{E}\Big[\int_0^{\delta_t} e^{-\beta s} f\big(t+s, \boldsymbol{X}_{t+s}^{\boldsymbol{u}}, \boldsymbol{u}\big)\,\mathrm{d}s \\
+ e^{-\beta\delta_t}\, J\big(t+\delta_t, \boldsymbol{X}_{t+\delta_t}^{\boldsymbol{u}};\pi\big) \;\big|\; \boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{x}\Big] =: I_1 + I_2.
\end{aligned} \tag{S.1.1}
$$

Since f is continuous at $(t,\boldsymbol{x})$ and $(\boldsymbol{X}_s^{\boldsymbol{u}})_{s\geq t}$ is càdlàg, by dominated convergence theorem, one has $\lim_{\delta_t\downarrow 0} \frac{1}{\delta_t}\mathbb{E}[\int_0^{\delta_t} e^{-\beta s}\, f(t+s, \boldsymbol{X}_{t+s}^{\boldsymbol{u}}, \boldsymbol{u})\,\mathrm{d}s \,|\boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{x}] = f(t,\boldsymbol{x},\boldsymbol{u})$, and hence

$$
I_1 = f(t,\boldsymbol{x},\boldsymbol{u})\,\delta_t + o(\delta_t), \quad (\delta_t \downarrow 0). \tag{S.1.2}
$$

For I_2, applying Itô's formula to $e^{-\beta(h-t)}J(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi)$ and integrating over $[t, t+\delta_t)$ gives $e^{-\beta\delta_t}J\big(t+\delta_t, \boldsymbol{X}_{t+\delta_t}^{\boldsymbol{u}};\pi\big) = J\big(t,\boldsymbol{X}_t^{\boldsymbol{u}};\pi\big) + \int_t^{t+\delta_t} e^{-\beta(h-t)}\big[\partial_t J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big) + \mathcal{L}^{\boldsymbol{u}} J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big) - \beta J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big)\big]\,\mathrm{d}h + \big(M_{t+\delta_t} - M_t\big)$, where $\mathcal{L}^{\boldsymbol{u}}$ is the generator associated with $\boldsymbol{u}$, and $(M_h)_{h\geq t}$ is a martingale with $\mathbb{E}[M_{t+\delta_t} - M_t|\boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{x}] = 0$. Thus,

$$
\begin{aligned}
I_2 = J(t,\boldsymbol{x};\pi) + \mathbb{E}\Big[\int_t^{t+\delta_t} e^{-\beta(h-t)}\big[\partial_t J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big) + \mathcal{L}^{\boldsymbol{u}} J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big) \\
- \beta J\big(h,\boldsymbol{X}_h^{\boldsymbol{u}};\pi\big)\big]\,\mathrm{d}h \;\big|\; \boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{x}\Big].
\end{aligned}
$$

By the regularity of J and $\boldsymbol{X}_s^{\boldsymbol{u}}$, as $\delta_t \to 0$ we obtain

$$
I_2 = J(t,\boldsymbol{x};\pi) + \big[\partial_t J(t,\boldsymbol{x};\pi) + \mathcal{L}^{\boldsymbol{u}} J(t,\boldsymbol{x};\pi) - \beta J(t,\boldsymbol{x};\pi)\big]\delta_t + o(\delta_t). \tag{S.1.3}
$$

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Combining (S.1.2) and (S.1.3), we conclude that

$$Q_{\delta_t}(t,\boldsymbol{x},\boldsymbol{u};\pi) = J(t,\boldsymbol{x};\pi) + \big[\partial_t J(t,\boldsymbol{x};\pi) + f(t,\boldsymbol{x},\boldsymbol{u}) + \mathcal{L}^{\boldsymbol{u}} J(t,\boldsymbol{x};\pi) - \beta J(t,\boldsymbol{x};\pi)\big]\delta_t + o(\delta_t).$$

Recall the Hamiltonian $\mathscr{H}(t,\boldsymbol{x},\boldsymbol{u};\pi) := f(t,\boldsymbol{x},\boldsymbol{u}) + \mathcal{L}^{\boldsymbol{u}} J(t,\boldsymbol{x};\pi)$ and the definition $q(t,\boldsymbol{x},\boldsymbol{u};\pi) := \lim_{\delta_t\downarrow 0} \frac{Q_{\delta_t}(t,\boldsymbol{x},\boldsymbol{u};\pi) - J(t,\boldsymbol{x};\pi)}{\delta_t}$, we achieve the desired expression (3.8).

S.1.2. Proof of Theorem 3.4. We first prove the performance-difference identity (3.11), following the idea of [2, Theorem 2] and adapting it to the present time-inhomogeneous jump-diffusion setting. We then apply this identity to a perturbed policy family and differentiate at the reference parameter. This yields the policy-gradient formula and completes the proof of Theorem 3.4.

To proceed, we first state the following identity, which follows [2, Lemma 9].

LEMMA S.1.1. *Let π and $\hat{\pi}$ be two stochastic policies, $J(\cdot,\cdot;\pi)$ be the value function under π. Let $\mathcal{L}^{\pi}$ denote the generator under policy π. Then, for all $(t,\boldsymbol{x})$,*

$$\begin{aligned}&\tilde{f}(t,\boldsymbol{x},\hat{\pi}) - \tilde{f}(t,\boldsymbol{x},\pi) + \big(\mathcal{L}^{\hat{\pi}} - \mathcal{L}^{\pi}\big) J(t,\boldsymbol{x};\pi)\\ &\quad = \int_{\mathcal{A}} \big(q(t,\boldsymbol{x},\boldsymbol{u};\pi) - \gamma \log \hat{\pi}(\boldsymbol{u}\mid t,\boldsymbol{x})\big)\,\hat{\pi}(\boldsymbol{u}\mid t,\boldsymbol{x})\mathrm{d}\boldsymbol{u}.\end{aligned} \tag{S.1.4}$$

Proof of performance-difference (3.11). Recall $\mu^{\hat{\pi},t,\boldsymbol{x}}$ for the discounted occupation measure induced by $\hat{\pi}$ starting from $(t,\boldsymbol{x})$. Apply Lemma 3.3 to $J(s,\boldsymbol{y};\pi)$ gives

$$\int_{[t,\infty)\times\mathbb{R}^d} \big(-\partial_s J(\cdot,\cdot;\pi) - \mathcal{L}^{\hat{\pi}} J(\cdot,\cdot;\pi) + \beta J(\cdot,\cdot;\pi)\big)(s,\boldsymbol{y})\,\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) = J(t,\boldsymbol{x};\pi). \tag{S.1.5}$$

On the other hand, since $J(\cdot,\cdot;\pi)$ is the value function under π, it satisfies (2.6). Integrating it against the measure $\mu^{\hat{\pi},t,\boldsymbol{x}}$ gives

$$0 = \int_{[t,\infty)\times\mathbb{R}^d} \big(\partial_s J(\cdot,\cdot;\pi) + \mathcal{L}^{\pi} J(\cdot,\cdot;\pi) + \tilde{f}(\cdot,\cdot;\pi) - \beta J(\cdot,\cdot;\pi)\big)(s,\boldsymbol{y})\,\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}). \tag{S.1.6}$$

Adding (S.1.6) to (S.1.5) produces

$$J(t,\boldsymbol{x};\pi) = \int_{[t,\infty)\times\mathbb{R}^d} \big[\tilde{f}(s,\boldsymbol{y};\pi) + \big(\mathcal{L}^{\pi} - \mathcal{L}^{\hat{\pi}}\big) J(s,\boldsymbol{y};\pi)\big]\,\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}). \tag{S.1.7}$$

By the definition of $J(t,\boldsymbol{x};\hat{\pi})$ and Lemma 3.2, we also have $J(t,\boldsymbol{x};\hat{\pi}) = \int_{[t,\infty)\times\mathbb{R}^d} \tilde{f}(s,\boldsymbol{y};\hat{\pi})\,\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y})$. Subtracting it from (S.1.7),

$$J(t,\boldsymbol{x};\hat{\pi}) - J(t,\boldsymbol{x};\pi) = \int_{[t,\infty)\times\mathbb{R}^d} \big(\tilde{f}(s,\boldsymbol{y};\hat{\pi}) - \tilde{f}(s,\boldsymbol{y};\pi) + \big(\mathcal{L}^{\hat{\pi}} - \mathcal{L}^{\pi}\big) J(s,\boldsymbol{y};\pi)\big)\,\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}). \tag{S.1.8}$$

By Lemma S.1.1,

$$J(t,\boldsymbol{x};\hat{\pi}) - J(t,\boldsymbol{x};\pi) = \int_{[t,\infty)\times\mathbb{R}^d}\int_{\mathcal{A}} \big(q(s,\boldsymbol{y},\boldsymbol{u};\pi) - \gamma\log\hat{\pi}(\boldsymbol{u}\mid s,\boldsymbol{y})\big)\,\hat{\pi}(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u}\mu^{\hat{\pi},t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}).$$

This proves the performance-difference formula.

We now prove the policy-gradient formula. Fix $(t,\boldsymbol{x}) \in [0,\infty)\times\mathbb{R}^d$ and a reference parameter θ_0. It suffices to prove for any h

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(t,\boldsymbol{x};\pi_{\theta_0+\varepsilon h})\big|_{\varepsilon=0} &= \int_{[t,\infty)\times\mathbb{R}^d}\int_{\mathcal{A}} \big\langle h, \nabla_\theta \log\pi_\theta(\boldsymbol{u}\mid s,\boldsymbol{y})\big|_{\theta=\theta_0}\big\rangle\\ &\quad \big(q(s,\boldsymbol{y},\boldsymbol{u};\pi_{\theta_0}) - \gamma\log\pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\big)\pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u}\mu^{\theta_0,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}),\end{aligned} \tag{S.1.9}$$

then normalizing $\beta\mu^{\theta_0,t,\boldsymbol{x}}$ yields the expectation form in the theorem. Apply the performance-difference formula with baseline π_{θ_0} and perturbed policy $\pi_{\theta_0+\varepsilon h}$ yields

$$J(t,\boldsymbol{x};\pi_{\theta_0+\varepsilon h}) - J(t,\boldsymbol{x};\pi_{\theta_0}) = \int_{[t,\infty)\times\mathbb{R}^d} \upsilon(\varepsilon;s,\boldsymbol{y})\,\mu^{\theta_0+\varepsilon h,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}), \tag{S.1.10}$$

where $\upsilon(\varepsilon;s,\boldsymbol{y})$ is defined as,

$$\upsilon(\varepsilon;s,\boldsymbol{y}) := \int_{\mathcal{A}} \pi_{\theta_0+\varepsilon h}(\boldsymbol{u}\mid s,\boldsymbol{y})\Big(q(s,\boldsymbol{y},\boldsymbol{u};\pi_{\theta_0}) - \gamma\log\pi_{\theta_0+\varepsilon h}(\boldsymbol{u}\mid s,\boldsymbol{y})\Big)\,\mathrm{d}\boldsymbol{u}. \tag{S.1.11}$$

Note that $\upsilon(0;s,\boldsymbol{y}) = 0$ for all $(s,\boldsymbol{y})$ due to the definition (3.8) of q and the PDE (2.6) satisfied by $J(\cdot,\cdot;\pi_{\theta_0})$. Consequently, add and subtract $\mu^{\theta_0,t,\boldsymbol{x}}$ from the right hand side of (S.1.10) gives:

$$\begin{aligned} J(t,\boldsymbol{x};\pi_{\theta_0+\varepsilon h}) - J(t,\boldsymbol{x};\pi_{\theta_0}) &= \int_{[t,\infty)\times\mathbb{R}^d} \big(\upsilon(\varepsilon;s,\boldsymbol{y}) - \upsilon(0;s,\boldsymbol{y})\big)\,\mu^{\theta_0+\varepsilon h,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) \\ &= \int_{[t,\infty)\times\mathbb{R}^d} \big(\upsilon(\varepsilon;s,\boldsymbol{y}) - \upsilon(0;s,\boldsymbol{y})\big)\,\mu^{\theta_0,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) \\ &\quad + \int_{[t,\infty)\times\mathbb{R}^d} \big(\upsilon(\varepsilon;s,\boldsymbol{y}) - \upsilon(0;s,\boldsymbol{y})\big)\,\big(\mu^{\theta_0+\varepsilon h,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) - \mu^{\theta_0,t,\boldsymbol{x}}(\mathrm{d}s,\mathrm{d}\boldsymbol{y})\big). \end{aligned} \tag{S.1.12}$$

For a direction h, write $\mu_\varepsilon := \mu^{\theta_0+\varepsilon h,t,\boldsymbol{x}}$, $\mu_0 := \mu^{\theta_0,t,\boldsymbol{x}}$, and define

$$g_{\varepsilon,h}(s,\boldsymbol{y},\boldsymbol{u}) := \pi_{\theta_0+\varepsilon h}(\boldsymbol{u}\mid s,\boldsymbol{y})(q(s,\boldsymbol{y},\boldsymbol{u};\pi_{\theta_0}) - \gamma\log\pi_{\theta_0+\varepsilon h}(\boldsymbol{u}\mid s,\boldsymbol{y})).$$

We impose the following local domination and continuity conditions. For every h, there exist $\varepsilon_0 > 0$ and a nonnegative measurable function $G_h(s,\boldsymbol{y},\boldsymbol{u})$ such that

$$\Big|\frac{\mathrm{d}}{\mathrm{d}\varepsilon} g_{\varepsilon,h}(s,\boldsymbol{y},\boldsymbol{u})\Big| \le G_h(s,\boldsymbol{y},\boldsymbol{u}), \qquad \forall|\varepsilon|\le\varepsilon_0 \text{ and a.e. } (s,\boldsymbol{y},\boldsymbol{u}).$$

Define $w_h(s,\boldsymbol{y}) := \int_{\mathcal{A}} G_h(s,\boldsymbol{y},\boldsymbol{u})\,\mathrm{d}\boldsymbol{u}$, and assume that $\int_{[t,\infty)\times\mathbb{R}^d} w_h(s,\boldsymbol{y})\,\mu_0(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) < \infty$. Finally, assume that

$$\|\mu_\varepsilon - \mu_0\|_{w_h} \to 0 \qquad \text{as } \varepsilon\to 0,$$

where, for a finite signed measure η, $\|\eta\|_{w_h} := \int_{[t,\infty)\times\mathbb{R}^d} w_h(s,\boldsymbol{y})\,|\eta|(\mathrm{d}s,\mathrm{d}\boldsymbol{y})$. Under the preceding assumptions, differentiation under the integral in (S.1.11) is justified:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\upsilon(\varepsilon;s,\boldsymbol{y})\big|_{\varepsilon=0} &= \int_{\mathcal{A}} \big\langle h, \nabla_\theta\pi_\theta(\boldsymbol{u}\mid s,\boldsymbol{y})\big|_{\theta=\theta_0}\big\rangle\big(q(s,y,\boldsymbol{u};\pi_{\theta_0}) - \gamma\log\pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\big)\,\mathrm{d}\boldsymbol{u} \\ &\quad - \gamma\int_{\mathcal{A}} \pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\big\langle h, \nabla_\theta\log\pi_\theta(\boldsymbol{u}\mid s,\boldsymbol{y})\big|_{\theta=\theta_0}\big\rangle\,\mathrm{d}\boldsymbol{u}. \end{aligned} \tag{S.1.13}$$

Using $\nabla_\theta\pi_\theta = \pi_\theta\nabla_\theta\log\pi_\theta$ and the normalization $\int_{\mathcal{A}}\pi_\theta(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} = 1$ (hence $\int_{\mathcal{A}}\nabla_\theta\pi_\theta(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} = 0$), the second term vanishes and (S.1.13) can be written as

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\upsilon(\varepsilon;s,\boldsymbol{y})\big|_{\varepsilon=0} &= \int_{\mathcal{A}} \big\langle h, \nabla_\theta\log\pi_\theta(\boldsymbol{u}\mid s,\boldsymbol{y})\big|_{\theta=\theta_0}\big\rangle \\ &\quad \big(q(s,y,\boldsymbol{u};\pi_{\theta_0}) - \gamma\log\pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\big)\pi_{\theta_0}(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u}. \end{aligned} \tag{S.1.14}$$

Now, define $D_\varepsilon(s,\boldsymbol{y}) := \frac{\upsilon(\varepsilon;s,\boldsymbol{y})-\upsilon(0;s,\boldsymbol{y})}{\varepsilon}$. By the mean-value theorem and the domination assumption, one has $|D_\varepsilon(s,\boldsymbol{y})| \le w_{\boldsymbol{h}}(s,\boldsymbol{y})$, $0<|\varepsilon|\le\varepsilon_0$, and $D_\varepsilon(s,\boldsymbol{y}) \to \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\upsilon(\varepsilon;s,\boldsymbol{y})\big|_{\varepsilon=0}$ pointwise. Then, by dominated convergence,

$$\lim_{\varepsilon\to 0}\int_{[t,\infty)\times\mathbb{R}^d} D_\varepsilon(s,\boldsymbol{y})\,\mu_0(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) = \int_{[t,\infty)\times\mathbb{R}^d} \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\upsilon(\varepsilon;s,\boldsymbol{y})\big|_{\varepsilon=0}\,\mu_0(\mathrm{d}s,\mathrm{d}\boldsymbol{y}).$$

Moreover,

$$\Big|\int_{[t,\infty)\times\mathbb{R}^d} D_\varepsilon(s,\boldsymbol{y})\,(\mu_\varepsilon-\mu_0)(\mathrm{d}s,\mathrm{d}\boldsymbol{y})\Big| \le \int_{[t,\infty)\times\mathbb{R}^d} w_h(s,\boldsymbol{y})\,|\mu_\varepsilon-\mu_0|(\mathrm{d}s,\mathrm{d}\boldsymbol{y}) = \|\mu_\varepsilon-\mu_0\|_{w_h}\to 0. \qquad \square$$

Dividing (S.1.12) by ε and letting $\varepsilon\to 0$ therefore gives (S.1.9). Since this holds for every direction h, it gives the Gâteaux derivative of $J(t,\boldsymbol{x};\pi_\theta)$ at θ_0. Using that $\beta\mu^{\theta_0,t,\boldsymbol{x}}$ is a probability measure then yields the policy-gradient representation (3.12).

S.1.3. Derivation of Lemma 3.5.

Proof. Fix $\delta_t>0, t\ge 0$, $\boldsymbol{x}\in\mathbb{R}^d$, and $\boldsymbol{u}\in\mathcal{A}$, and recall $(\boldsymbol{X}_s^{\boldsymbol{u}})_{s\ge t}$ from Section 3.2.1. Apply the Itô-Lévy formula to $e^{-\beta(s-t)}J(s,\boldsymbol{X}_s^{\boldsymbol{u}};\pi)$ on $[t,t+\delta_t)$, we obtain

$$\begin{aligned} &e^{-\beta\delta_t}J\big(t+\delta_t,\boldsymbol{X}_{t+\delta_t}^{\boldsymbol{u}};\pi\big)-J\big(t,\boldsymbol{X}_t^{\boldsymbol{u}};\pi\big)\\ &\quad=\int_t^{t+\delta_t} e^{-\beta(s-t)}\big(\partial_s J+\mathcal{L}^{\boldsymbol{u}}J-\beta J\big)(s,\boldsymbol{X}_s^{\boldsymbol{u}};\pi)\,\mathrm{d}s\;+\;\tilde{M}_{t+\delta_t}-\tilde{M}_t, \end{aligned} \tag{S.1.15}$$

where $\tilde{M}$ collects the stochastic integrals with respect to the Brownian motion and the compensated Poisson random measure. Taking conditional expectation with respect to $\mathcal{F}_t$ and using the integrability assumption in Lemma 3.5 (which ensures the local martingale term is a true martingale on $[t,t+\delta_t)$), we get

$$\begin{aligned} &\mathbb{E}\big[e^{-\beta\delta_t}J\big(t+\delta_t,\boldsymbol{X}_{t+\delta_t}^{\boldsymbol{u}};\pi\big)-J\big(t,\boldsymbol{X}_t^{\boldsymbol{u}};\pi\big)\,\big|\,\mathcal{F}_t\big]\\ &\quad=\mathbb{E}\Big[\int_t^{t+\delta_t} e^{-\beta(s-t)}\big(\partial_s J+\mathcal{L}^{\boldsymbol{u}}J-\beta J\big)(s,\boldsymbol{X}_s^{\boldsymbol{u}};\pi)\,\mathrm{d}s\,\Big|\,\mathcal{F}_t\Big]. \end{aligned} \tag{S.1.16}$$

By continuity of J and the coefficients, together with dominated convergence (using the integrability assumption in Lemma 3.5), as $\delta_t\to 0$,

$$\mathbb{E}\Big[\textstyle\int_t^{t+\delta_t} e^{-\beta(s-t)}\big(\partial_s J+\mathcal{L}^{\boldsymbol{u}}J-\beta J\big)(s,\boldsymbol{X}_s^{\boldsymbol{u}};\pi)\,\mathrm{d}s\,\Big|\,\mathcal{F}_t\Big]=\big(\partial_t J+\mathcal{L}^{\boldsymbol{u}}J-\beta J\big)(t,\boldsymbol{X}_t^{\boldsymbol{u}};\pi)\,\delta_t+o(\delta_t).$$

Since $f(t,\boldsymbol{X}_t^{\boldsymbol{u}},\boldsymbol{u})$ is $\mathcal{F}_t$-measurable, one has

$$\begin{aligned} &\mathbb{E}\big[f(t,\boldsymbol{X}_t^{\boldsymbol{u}},\boldsymbol{u})\,\delta_t+e^{-\beta\delta_t}J\big(t+\delta_t,\boldsymbol{X}_{t+\delta_t}^{\boldsymbol{u}};\pi\big)-J\big(t,\boldsymbol{X}_t^{\boldsymbol{u}};\pi\big)|\mathcal{F}_t\big]\\ &\quad=\big(\partial_t J+\mathcal{L}^{\boldsymbol{u}}J-\beta J+f\big)(t,\boldsymbol{X}_t^{\boldsymbol{u}},\boldsymbol{u};\pi)\,\delta_t+o(\delta_t). \end{aligned} \tag{S.1.17}$$

Dividing it by δ_t and using (3.14) and the definition (3.8) of q-function gives (3.15). $\square$

S.1.4. Proof for (4.10).

Proof. Let $\mathscr{H}(u):=\mathscr{H}(x,u,\nabla_x V,\nabla_x^2 V)$, $Z(x):=\int_{\mathcal{A}}\exp\Big(\frac{1}{\gamma}\mathscr{H}(\tilde{u})\Big)\,d\tilde{u}$. By the Gibbs representation of the optimal policy, $\pi^*(u\mid x)=\frac{\exp\left(\frac{1}{\gamma}\mathscr{H}(u)\right)}{Z(x)}$. Hence $\log\pi^*(u\mid x)=\frac{1}{\gamma}\mathscr{H}(u)-\log Z(x)$, which implies $\mathscr{H}(u)-\gamma\log\pi^*(u\mid x)=\gamma\log Z(x)$. For the entropy-regularized HJB equation,

$$\beta V(x)=\int_{\mathcal{A}}\big(\mathscr{H}(u)-\gamma\log\pi^*(u\mid x)\big)\pi^*(u\mid x)\,du\,,$$

substituting π^* into HJB and using the normalization $\int_{\mathcal{A}}\pi^*(u\mid x)\,du=1$ yields

$$\beta V(x)=\gamma\log Z(x)=\gamma\log\int_{\mathcal{A}}\exp\left(\frac{1}{\gamma}\mathscr{H}(x,u,\nabla_x V,\nabla_x^2 V)\right)\,du.$$

which is exactly (4.10). $\square$

S.2. Proof For Benchmarks.

S.2.1. Closed-form Solution in the LQ Case. For the linear-quadratic problem, we adopt the quadratic ansatz

$$V(t,\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{H}(t)\boldsymbol{x} + g_\gamma(t), \qquad \boldsymbol{H}(t) \in \mathbb{S}^d, \quad g_\gamma(t) \in \mathbb{R}. \tag{S.2.1}$$

We show that, under this ansatz, the entropy-regularized HJB admits an explicit Gaussian optimizer and reduces to a Riccati–scalar ODE system.

Proof. Recall that $f(t,\boldsymbol{x},\boldsymbol{u}) = -(\boldsymbol{u}^\top \boldsymbol{R}(t)\boldsymbol{u} + \boldsymbol{x}^\top \boldsymbol{Q}(t)\boldsymbol{x})$, $\boldsymbol{R}(t) \succ \boldsymbol{0}$, $\boldsymbol{Q}(t) \succeq \boldsymbol{0}$. Substituting (S.2.1) into the entropy-regularized HJB, the policy-dependent part becomes

$$\sup_{\pi(\cdot|t,\boldsymbol{x})} \left\{ \mathbb{E}_\pi[-\boldsymbol{u}^\top \boldsymbol{R}(t)\boldsymbol{u} + 2\,\boldsymbol{u}^\top \boldsymbol{B}(t)^\top \boldsymbol{H}(t)\boldsymbol{x}] + \gamma\, \mathbb{E}_\pi[-\log \pi(\boldsymbol{u} \mid t,\boldsymbol{x})] \right\},$$

By the optimizer is of Gibbs form, $\pi^*(\boldsymbol{u} \mid t,\boldsymbol{x}) \propto \exp\Big(\frac{\mathscr{H}(t,\boldsymbol{x},\boldsymbol{u})}{\gamma}\Big) =$
$\mathcal{N}\big(\boldsymbol{R}(t)^{-1}\boldsymbol{B}(t)^\top \boldsymbol{H}(t)\boldsymbol{x},\, \frac{\gamma}{2}\boldsymbol{R}(t)^{-1}\big)$, and $\mathbb{E}_{\pi^*}[\boldsymbol{u} \mid t,\boldsymbol{x}] = \boldsymbol{R}(t)^{-1}\boldsymbol{B}(t)^\top \boldsymbol{H}(t)\boldsymbol{x}$,
$\mathrm{Var}_{\pi^*}(\boldsymbol{u} \mid t,\boldsymbol{x}) = \frac{\gamma}{2}\boldsymbol{R}(t)^{-1}$, matching the coefficients in HJB, we obtain

$$\boldsymbol{H}'(t) = \beta \boldsymbol{H}(t) + \boldsymbol{Q}(t) - \boldsymbol{H}(t)\boldsymbol{B}(t)\boldsymbol{R}(t)^{-1}\boldsymbol{B}(t)^\top \boldsymbol{H}(t), \tag{S.2.2}$$

and

$$g'_\gamma(t) = \beta g_\gamma(t) - \mathrm{Tr}\big(\boldsymbol{\Sigma}(t)\boldsymbol{\Sigma}(t)^\top \boldsymbol{H}(t)\big) - \mathrm{Tr}\big(\boldsymbol{\Lambda}(t)\ \mathrm{diag}(\boldsymbol{\alpha}(t))\, \boldsymbol{H}(t)\ \mathrm{diag}(\boldsymbol{\alpha}(t))\big) - c_\gamma(t), \tag{S.2.3}$$

where

$$c_\gamma(t) = \frac{\gamma}{2}\big(m \log(\pi\gamma) - \log\det \boldsymbol{R}(t)\big), \qquad \boldsymbol{\Lambda}(t) := \mathrm{diag}(\lambda_i(t)).$$

Hence the value function is obtained.

In the standard case $\gamma = 0$, the entropy term disappears and the optimal stochastic policy collapses to the deterministic feedback control

$$\boldsymbol{u}^*(t,\boldsymbol{x}) = \boldsymbol{R}(t)^{-1}\boldsymbol{B}(t)^\top \boldsymbol{H}(t)\boldsymbol{x}, \tag{S.2.4}$$

which is exactly the mean of optimal policy when $\gamma > 0$, and it can be obtained directly from the first-order optimality condition. Accordingly, (S.2.2) remains unchanged, while (S.2.3) reduces to the same scalar equation without the entropy correction term $c_\gamma(t)$.

Finally, the ODE system above determines the candidate solution, and the admissible branch is selected by the coefficient class. If the coefficients converge as $t \to \infty$, we impose

$$\lim_{t\to\infty} \boldsymbol{H}(t) = \boldsymbol{H}_\infty, \qquad \lim_{t\to\infty} g_\gamma(t) = g_\infty^\gamma,$$

where $\boldsymbol{H}_\infty$ solves the limiting algebraic Riccati equation and g_∞^γ is determined by the corresponding stationary scalar balance. To implement this asymptotic condition in the one-dimensional setting, we set $\boldsymbol{H}(T_\infty) = \boldsymbol{H}_\infty < 0$, choosing the stabilizing solution of the limiting algebraic Riccati equation, and then integrate the Riccati equation backward. For the parameter values used in our experiments, the computed solution satisfies $\boldsymbol{H}(t) \prec 0$ at every time-grid point. If the coefficients are P-periodic, then we impose the periodic boundary condition

$$\boldsymbol{H}(t+P) = \boldsymbol{H}(t), \qquad g_\gamma(t+P) = g_\gamma(t).$$

Numerically, the value of $\boldsymbol{H}(P) = \boldsymbol{H}(0)$ is determined by a shooting method. For the parameters used, the resulting periodic solution satisfies $\boldsymbol{H}(t) \prec 0$ throughout $[0,P]$.

In the time-homogeneous case, $\boldsymbol{H}'(t) = 0$ and $g'_\gamma(t) = 0$, so the system reduces to the associated algebraic Riccati equation together with the stationary scalar equation. ∎

S.2.2. Closed-form Solution of Multi-agent Game.

Proof. Recall that $Y_t^i := \frac{1}{n}\sum_{j\neq i} X_t^j$ represents the average wealth of other agents, and that $V^i(t,x,y)$ is defined in (4.14). Then Y_t^i satisfies

$$\mathrm{d}Y_t^i = \widehat{ub}\,\mathrm{d}t + \widehat{u\sigma}\,\mathrm{d}W_t^0 + \frac{1}{n}\sum_{j\neq i} u_j\eta_j\,\mathrm{d}W_t^j + \frac{1}{n}\sum_{j\neq i} u_j\alpha_j\,\mathrm{d}M_t^j + \widehat{u\xi}\,\mathrm{d}M_t^0, \tag{S.2.5}$$

where $\widehat{ub} := \frac{1}{n}\sum_{j\neq i} u_j b_j$, $\widehat{u\sigma} := \frac{1}{n}\sum_{j\neq i} u_j\sigma_j$, $\widehat{u^2\eta^2} := \frac{1}{n}\sum_{j\neq i} u_j^2\eta_j^2$, $\widehat{u\xi} := \frac{1}{n}\sum_{j\neq i} u_j\xi_j$. By the dynamic programming principle, V^i satisfies the HJB equation

$$\partial_t V^i + \sup_{u_i\in\mathbb{R}}\big\{\mathcal{L}^{i,u_i}V^i + f_i\big\} - \beta V^i = 0, \tag{S.2.6}$$

where $\mathcal{L}^{i,u_i}$ is the generator of the pair (X_t^i, Y_t^i) under the control u_i for agent i, with the controls u_j, $j\neq i$ held fixed. A direct computation yields

(S.2.7)

$$\begin{aligned}
&\mathcal{L}^{i,u_i}V^i(t,x,y)\\
&= \big[u_i b_i\,V_x^i + \widehat{ub}\,V_y^i + \tfrac{1}{2}(\eta_i^2+\sigma_i^2)u_i^2\,V_{xx}^i + \tfrac{1}{2}\big((\widehat{u\sigma})^2 + \tfrac{1}{n}\widehat{u^2\eta^2}\big)V_{yy}^i + u_i\sigma_i\widehat{u\sigma}\,V_{xy}^i\big]_{(t,x,y)}\\
&\quad + \lambda_i\big(V^i(t,x+u_i\alpha_i,y) - V^i(t,x,y) - u_i\alpha_i\,V_x^i(t,x,y)\big)\\
&\quad + \sum_{j\neq i}\lambda_j\big(V^i\big(t,x,y+\tfrac{u_j\alpha_j}{n}\big) - V^i(t,x,y) - \tfrac{u_j\alpha_j}{n}\,V_y^i(t,x,y)\big)
\end{aligned}$$

(S.2.8)

$$+ \lambda_0\big(V^i(t,x+u_i\xi_i,y+\widehat{u\xi}) - V^i(t,x,y) - u_i\xi_i\,V_x^i(t,x,y) - \widehat{u\xi}\,V_y^i(t,x,y)\big).$$

We seek solutions of the form

$$V^i(t,x,y) = -\frac{1}{K_i(t)}\exp\big(-\frac{1}{\varrho_i}\big((1-\frac{\varpi_i}{n})x - \varpi_i y\big)\big). \tag{S.2.9}$$

Define $\chi_i = \frac{1-\varpi_i/n}{\varrho_i}$, and $\rho_i = \frac{\varpi_i}{\varrho_i}$. Substituting the ansatz into (S.2.8) and dividing by $V^i<0$, the u_i-dependent terms from the drift and diffusion are $\frac{1}{V^i}\Big(u_i b_i V_x^i + \frac{1}{2}(\eta_i^2+\sigma_i^2)u_i^2 V_{xx}^i + u_i\sigma_i\widehat{u\sigma}\,V_{xy}^i\Big) = -\chi_i b_i u_i + \frac{1}{2}\chi_i^2(\eta_i^2+\sigma_i^2)u_i^2 - \chi_i\rho_i\sigma_i\widehat{u\sigma}\,u_i$. The jump terms contribute $\frac{\lambda_i}{V^i}\big(V^i(t,x+u_i\alpha_i,y) - V^i(t,x,y) - u_i\alpha_i V_x^i\big) = \lambda_i\big(e^{-\chi_i\alpha_i u_i} - 1 + \chi_i\alpha_i u_i\big)$, and $\frac{\lambda_0}{V^i}\big(V^i(t,x+u_i\xi_i,y+\widehat{u\xi}) - V^i(t,x,y) - u_i\xi_i V_x^i - \widehat{u\xi}V_y^i\big) = \lambda_0\big(e^{-\chi_i\xi_i u_i+\rho_i\widehat{u\xi}} - 1 + \chi_i\xi_i u_i - \rho_i\widehat{u\xi}\big)$. Collecting all u_i-dependent terms defines the function Ψ_i:

$$\begin{aligned}
\Psi_i(u) = &-\chi_i b_i u + \frac{1}{2}\chi_i^2(\eta_i^2+\sigma_i^2)u^2 - \chi_i\rho_i\sigma_i\widehat{u\sigma}\,u + \lambda_i\big(e^{-\chi_i\alpha_i u} - 1 + \chi_i\alpha_i u\big)\\
&+ \lambda_0\Big(e^{-\chi_i\xi_i u+\rho_i\widehat{u\xi}} + \chi_i\xi_i u\Big).
\end{aligned} \tag{S.2.10}$$

The first order condition $\Psi_i'(u)=0$ gives a candidate optimal control for agent i. Since Ψ_i is strictly convex in u, its unique minimizer u_i^* is characterized by $\Psi_i'(u_i^*)=0$. Because the u-dependent term in agent i's HJB equation is $V_i\Psi_i(u)$ and $V_i<0$, minimizing Ψ_i is equivalent to maximizing the HJB expression over u. Consequently, a collection of feedback controls $\boldsymbol{u}^* = (u_1^*,\dots,u_n^*)$ forms a Markovian Nash equilibrium

if and only if it solves the coupled system $\Psi_i'(u_i^*) = 0$, $i = 1, 2, \ldots, n$. This proves the equilibrium characterization.

Matching the coefficients and using time homogeneity gives $\Psi_i(u_i^*) + K_i(t) + C_i - \beta = 0$, where C_i is independent of u_i and given by

$$\begin{aligned} C_i = {} & \rho_i\, \widehat{ub} + \frac{1}{2}\,\rho_i^2\Big(\frac{1}{n^2}\sum_{j\neq i}(u_j\eta_j)^2 + \Big(\frac{1}{n}\sum_{j\neq i}u_j\sigma_j\Big)^2\Big) \\ & + \sum_{j\neq i}\lambda_j\Big(\exp\Big(\frac{\varpi_i}{\varrho_i}\,\frac{u_j\alpha_j}{n}\Big) - 1 - \frac{\varpi_i}{\varrho_i}\,\frac{u_j\alpha_j}{n}\Big) - \lambda_0\,\rho_i\,\widehat{u\xi}. \end{aligned} \tag{S.2.11}$$

Hence $K_i^* = \beta - \Psi_i(u_i^*) - C_i$, and the value function of agent i is

$$V^i(x,y) = -\frac{1}{\beta - \Lambda_i^*}\exp\left(-\frac{1}{\varrho_i}\Big(\big(1 - \tfrac{\varpi_i}{n}\big)x - \varpi_i y\Big)\right), \tag{S.2.12}$$

where $\Lambda_i^* := \Psi_i(u_i^*) + C_i$, and $\beta > \Lambda_i^*$ ensures the concavity. ∎

S.3. More Details For Experiments.

S.3.1. Numerical Simulation of State Dynamics. In all experiments we simulate the controlled jump-diffusion dynamics by an Euler–Maruyama scheme on a uniform grid $0 = t_0 < \cdots < t_K$ with $t_k := k\delta_t$. We do not directly discretize the averaged coefficients of the exploratory SDE (2.2). Instead, at each grid point t_k, we sample actions $\boldsymbol{u}_k^{(\ell)} \sim \pi_\theta(\cdot \mid t_k, \boldsymbol{X}_k^{(\ell)})$, hold this action fixed over $[t_k, t_{k+1})$, and evolve the resulting strict controlled jump-diffusion. Under the assumptions of [3], this grid-sampled process converges weakly to the exploratory SDE (2.2) as $\delta_t \downarrow 0$.

At each t_k, we sample an action $\boldsymbol{u}_k^{(\ell)} \sim \pi_\theta(\cdot \mid t_k, \boldsymbol{X}_k^{(\ell)})$, a Brownian increment $\Delta\boldsymbol{W}_k^{(\ell)} \sim \mathcal{N}(\boldsymbol{0}, \delta_t \boldsymbol{I}_d)$, and, under the Poisson jump specification, jump counts $\Delta N_k^{(i,\ell)} \sim \mathrm{Poisson}(\lambda_i\delta_t)$ independently across i, k, ℓ, and update

$$\begin{aligned} \boldsymbol{X}_{k+1}^{(\ell)} = {} & \boldsymbol{X}_k^{(\ell)} + \boldsymbol{b}(t_k, \boldsymbol{X}_k^{(\ell)}, \boldsymbol{u}_k^{(\ell)})\,\delta_t + \boldsymbol{\sigma}(t_k, \boldsymbol{X}_k^{(\ell)}, \boldsymbol{u}_k^{(\ell)})\,\Delta\boldsymbol{W}_k^{(\ell)} \\ & + \sum_{i=1}^{d}\boldsymbol{\alpha}(t_k, \boldsymbol{X}_k^{(\ell)}, \boldsymbol{u}_k^{(\ell)}, \boldsymbol{e}_i)\,\Delta M_k^{(i,\ell)}, \end{aligned} \tag{S.3.1}$$

where $\Delta M_k^{(i,\ell)} := \Delta N_k^{(i,\ell)} - \lambda_i\delta_t$ denotes the compensated jump increments. These realized Brownian increments $\Delta\boldsymbol{W}_k^{(\ell)}$ and jumps are also used in the martingale-corrected TD error in the main text.

S.3.2. Discretization of Non-local Term. We provide here the discretization of the martingale correction $\mathcal{I}^{\pi}_{t,t+\delta_t}$ defined in (3.3). This correction is subtracted from the one-step standard TD error to obtain the martingale-corrected critic update. For notational simplicity, we describe the discretization along one trajectory over the interval $[t_k, t_{k+1}]$.

The Brownian component on $[t_k, t_{k+1}]$ is approximated by $\big(\nabla_{\boldsymbol{x}} V_\psi(t_k, \boldsymbol{X}_k)^\top \boldsymbol{\sigma}(t_k, \boldsymbol{X}_k, \boldsymbol{u}_k)\big)\Delta\boldsymbol{W}_k$. The jump component is

$$\mathcal{I}_k^J := \int_{t_k}^{t_{k+1}}\int_{\mathbb{R}^d} e^{-\beta(s-t_k)}\Delta_{\boldsymbol{z}} V_\psi(s, \boldsymbol{X}_{s-}, \boldsymbol{u}_s)\,\widetilde{N}(\mathrm{d}s, \mathrm{d}\boldsymbol{z}),$$

where $\Delta_{\boldsymbol{z}} V_\psi(t, \boldsymbol{x}, \boldsymbol{u}) := V_\psi\big(t, \boldsymbol{x} + \boldsymbol{\alpha}(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{z})\big) - V_\psi(t, \boldsymbol{x})$. For a finite-activity jump

measure, using $\widetilde{N}(\mathrm{d}s, \mathrm{d}\boldsymbol{z}) = N(\mathrm{d}s, \mathrm{d}\boldsymbol{z}) - \nu(\mathrm{d}\boldsymbol{z})\,\mathrm{d}s$, we obtain

(S.3.2)

$$\mathcal{I}_k^J = \sum_{\tau_i \in (t_k, t_{k+1}]} e^{-\beta(\tau_i - t_k)} \Delta_{\boldsymbol{z}_i} V_\psi(\tau_i, \boldsymbol{X}_{\tau_i -}, \boldsymbol{u}_{\tau_i}) - \int_{t_k}^{t_{k+1}} e^{-\beta(s-t_k)} V_{\mathrm{non}}(s, \boldsymbol{X}_{s-}, \boldsymbol{u}_s)\,\mathrm{d}s,$$

where τ_i and $\boldsymbol{z}_i$ denote the realized jump times and jump marks, respectively, and

$$V_{\mathrm{non}}(t, \boldsymbol{x}, \boldsymbol{u}) := \int_{\mathbb{R}^d} \Delta_{\boldsymbol{z}} V_\psi(t, \boldsymbol{x}, \boldsymbol{u})\,\nu(\mathrm{d}\boldsymbol{z}) \tag{S.3.3}$$

is the instantaneous compensator rate of the jump increments of the critic. Applying the same left-endpoint approximation to (S.3.2) gives

$$\mathcal{I}_k^J \approx \sum_{i=N_{t_k}+1}^{N_{t_{k+1}}} \left[V_\psi\big(t_k, \boldsymbol{X}_k + \boldsymbol{\alpha}(t_k, \boldsymbol{X}_k, \boldsymbol{u}_k, \boldsymbol{z}_i)\big) - V_\psi(t_k, \boldsymbol{X}_k)\right] - \delta_t V_{\mathrm{non}}(t_k, \boldsymbol{X}_k, \boldsymbol{u}_k).$$

For the discrete Lévy measure used in our numerical experiments,

$$\nu(\mathrm{d}\boldsymbol{z}) = \sum_{i=1}^{d} \lambda_i \delta_{\boldsymbol{e}_i}(\mathrm{d}\boldsymbol{z}),$$

the compensator rate in (S.3.3) can be evaluated exactly as

$$V_{\mathrm{non}}(t, \boldsymbol{x}, \boldsymbol{u}) = \sum_{i=1}^{d} \lambda_i \left[V_\psi\big(t, \boldsymbol{x} + \boldsymbol{\alpha}(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{e}_i)\big) - V_\psi(t, \boldsymbol{x})\right]. \tag{S.3.4}$$

More generally, suppose that ν is known and has finite total mass $\lambda_\nu := \nu(\mathbb{R}^d) < \infty$, and define $\overline{\nu}(\mathrm{d}\boldsymbol{z}) := \nu(\mathrm{d}\boldsymbol{z})/\lambda_\nu$. Then

$$V_{\mathrm{non}}(t, \boldsymbol{x}, \boldsymbol{u}) = \lambda_\nu \mathbb{E}_{\boldsymbol{Z} \sim \overline{\nu}} \left[\Delta_{\boldsymbol{Z}} V_\psi(t, \boldsymbol{x}, \boldsymbol{u})\right].$$

Given independent samples $\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_M \sim \overline{\nu}$, it may be estimated by

$$\frac{\lambda_\nu}{M} \sum_{j=1}^{M} \left[V_\psi\big(t, \boldsymbol{x} + \boldsymbol{\alpha}(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{Z}_j)\big) - V_\psi(t, \boldsymbol{x})\right].$$

Alternatively, when the jump-mark dimension is sufficiently small, the integral in (S.3.3) may be evaluated by numerical quadrature.

Alternatively, one may introduce an auxiliary neural network $V_{\mathrm{non},\eta}(t, \boldsymbol{x}, \boldsymbol{u})$ to approximate the compensator rate. Since the corresponding compensated jump integral is a martingale, the network may be trained through the conditional moment condition

$$\mathbb{E}\Big[\sum_{\tau_i \in (t_k, t_{k+1}]} \Delta_{\boldsymbol{z}_i} V_\psi(t_k, \boldsymbol{X}_k, \boldsymbol{u}_k) - \delta_t V_{\mathrm{non},\eta}(t_k, \boldsymbol{X}_k, \boldsymbol{u}_k) \mid \mathcal{F}_{t_k}\Big] \approx 0,$$

In practice, this condition can be enforced by minimizing the empirical squared residual over sampled trajectories. The resulting approximation can then be substituted for V_{non} in the martingale-corrected TD error, thereby avoiding repeated evaluation of the nonlocal integral. A rigorous justification is beyond the scope of this work.

We also emphasize that V_{non} is needed only for the optional martingale-corrected TD update and is not required for the standard TD critic update.

S.3.3. Numerical Comparison of Usual and Martingale-Corrected TD Errors. To examine the variance-reduction effect of the martingale correction, we compare the usual TD error, implemented with the semi-gradient convention described in the main text, with its martingale-corrected counterpart on the five-dimensional jump-diffusion LQ problem. Since the martingale-corrected TD error requires the realized Brownian and jump increments, the purpose of this experiment is not to provide an information-matched comparison between two RL algorithms, but rather to isolate the effect of subtracting the martingale fluctuations.

We consider two experiments. First, fixing the analytic LQ solution, we evaluate both the usual and martingale-corrected TD residuals on the same states and using the same Brownian and Poisson increments. For each δ_t, we generate 30 independent minibatches with $L = 1{,}000$ trajectories. The ratio $\frac{\mathrm{Var}(\tilde{\delta}_{\mathrm{TD}})}{\mathrm{Var}(\delta_{\mathrm{TD}})}$ decreases from 5.35×10^{-4} at $\delta_t = 0.02$ to 9.34×10^{-6} at $\delta_t = 0.002$. We observe a similar reduction in the variability of the corresponding squared one-step losses. This directly confirms the variance-reduction effect of subtracting the martingale increments.

Second, we compare the two critic losses within the complete actor–critic training procedure. All algorithmic settings are kept fixed, and only the critic loss is changed from the usual TD error to the martingale-corrected TD error. We perform the complete training with three paired random seeds, 2025, 2026, and 42. This comparison follows the diffusion-only study of Zhou, Han, and Lu [**?**], which examines the effect of martingale correction on value-function and policy errors. Our jump-diffusion setting additionally includes the compensated Poisson martingale term. For each trained model, we evaluate the learned value function and policy on $L_{\mathrm{eval}} = 500$ common reference trajectories at $K + 1 = 501$ time points. At time t_k, the value-function error is

$$\mathrm{MSE}_V(t_k) = \frac{1}{L_{\mathrm{eval}}} \sum_{\ell=1}^{L_{\mathrm{eval}}} \left[\widehat{V}(t_k, \boldsymbol{X}_k^{(\ell)}) - V(t_k, \boldsymbol{X}_k^{(\ell)})\right]^2,$$

while the policy error is measured by the root-mean-square Gaussian W_2 distance

$$D_\pi^{W_2}(t_k) = \left\{ \frac{1}{L_{\mathrm{eval}}} \sum_{\ell=1}^{L_{\mathrm{eval}}} \left(\left\|\widehat{\boldsymbol{\mu}}_\ell(t_k) - \bar{\boldsymbol{\mu}}_\ell^*(t_k)\right\|_2^2 + \left\|\widehat{\mathrm{Std}}_\ell(t_k) - \mathrm{Std}_\ell^*(t_k)\right\|_2^2 \right) \right\}^{1/2}.$$

Here, $\overline{\mathrm{MSE}}_V$ and $\overline{D}_\pi^{W_2}$ denote the arithmetic averages of $\mathrm{MSE}_V(t_k)$ and $D_\pi^{W_2}(t_k)$, respectively, over all 501 time points. Following Eq. (4.3) under $d = m = 5$ and $T = 1$, we use the same parameter setting in Fig. 1 for the following tests.

Table SM1 reports the time-averaged value-function and policy errors. Each entry in Table SM1 is the mean ± sample standard deviation across the three seed-level results ($n = 3$); each seed-level result itself is evaluated using the same 500 reference trajectories. The time-averaged value-function MSE is reduced by 99.07%, while the time-averaged Gaussian policy-distribution distance is reduced by 74.10%, which indicates that the martingale-corrected TD update yields substantially smaller errors in both the learned value function and the policy distribution.

In addition, Figure SM1 shows the pointwise value-function and policy errors over all 501 evaluation time points. The curves represent the across-seed mean errors, while the shaded bands are constructed from the three seed-level values at each time point, with the lower and upper boundaries given by their minimum and maximum, respectively. For these three paired seeds, the results illustrate that reducing the variance of the critic learning signal can improve the accuracy of the learned value function and policy.

Table SM1: Time-averaged value-function MSE and Gaussian policy-distribution distance.

Critic loss	$\overline{\mathrm{MSE}}_V$	$\overline{D}_\pi^{W_2}$
Usual TD error	$(4.2792 \pm 4.0007) \times 10^{-1}$	$(3.8409 \pm 0.1306) \times 10^{-2}$
Martingale-corrected TD error	$(3.9824 \pm 3.2527) \times 10^{-3}$	$(9.9493 \pm 2.3502) \times 10^{-3}$

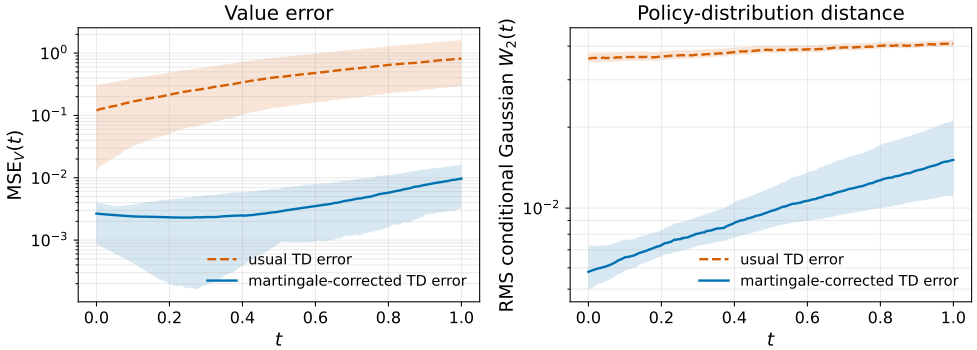


Fig. SM1: Comparison of the usual TD error and martingale-corrected TD error over three training seeds. Left: value-function MSE. Right: Gaussian policy-distribution W_2 distance. At each of the 501 time points, each seed-level metric is computed over 500 common reference trajectories; the curves show the mean across the three seeds, and the shaded bands span the minimum to maximum across these three seed-level values.

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